

REMOTE IT INFRASTRUCTURE CONSOLIDATION

Fortune 1000 CIOs are making the strategic choice to consolidate remote site IT infrastructure into central datacentre. They are compelled to move some or all remote file servers, email servers, backup, and other servers because through such site consolidation they can jointly address the need to reduce remote site operating costs and mandates for more rigorous security and compliance.

The stumbling block to consolidation, however, is the severe impact on application performance as seen by remote users. Relocating local servers to a datacentre and connecting them across a wide area network (WAN) link often results in order-of-magnitude slowdowns to response times and data transfer rates. At these levels of delay business processes are impacted forcing site consolidation efforts to be stalled.

CIOs often discover that upgrading bandwidth to remote sites has little or no effect on application performance. The problem lies instead with the way the applications interact with the server across the WAN. Microsoft Windows file systems, Microsoft Exchange, NAS, backup applications, CAD applications, and many others were developed with the idea that the client and server were local. Across the WAN, however, where congestion, resource contention, diverse routing conditions, and high latencies exist, these applications grind to a crawl.

Why Consolidate?

Remote site server consolidation is a clear win in terms of reducing operating costs and improving data security. However, there were compelling reasons for distributing server infrastructure in the first place.

The reason many companies chose to place servers at remote sites has been to deliver consistent application performance to remote users working with local data sets. Microsoft Exchange servers, for example, have commonly been deployed at remote sites with only 20-30 users because above those numbers, most Exchange messages end up being between local users. Provisioning of servers at remote sites, however, often leads to low resource utilisation and high costs. Since Exchange servers are typically resourced for a capacity of several thousand users, deploying a dedicated server for a few dozen means inefficient use of server resources. This same issue exists for file servers, and web servers. Worse, all those servers have to be managed, backed up, repaired, and patched.

Site consolidation has a tremendous ROI, as long as user application performance is preserved

Centralising servers at a data center means greater resource utilisation, and fewer servers to backup and patch. Since complexity is reduced, such consolidation also means lower IT staff requirements, less chance for errors, and better system security.

Because of the clear benefits, companies are trying to consolidate infrastructure as much as possible, yet many are surprised at how difficult it is to actually complete a successful site consolidation project. They find they can't deliver consistent end-to-end application performance even with significant upgrades to WAN bandwidth.

Three Barriers To Site Consolidation

When WANs are involved, client-server applications that worked fantastically on LANs break down and work poorly, or not at all. The reason is threefold: Constrained WAN bandwidth; TCP throughput drop-off with latency; Application chattiness multiplies the effect of latency.

Constrained WAN Bandwidth: WAN bandwidth is often orders of magnitude less than local area network (LAN) bandwidth. A typical remote office has between 64 kbps and T1 bandwidth (1.544 Mbps) or E1 (2 Mbps). Compared to modern LANs that have 100 Mbps to 1,000 Mbps, a remote site typically relies on less than 1 percent of the bandwidth to access data as a result of server consolidation.

From a pure bits-per-second perspective it's easy to see why moving a large file across a WAN link should take more time than over the LAN. However, it's often the other two constraints, not bandwidth, that result in low application performance.

TCP throughput drops-off with latency: All applications rely on underlying communications protocols which, for reliable transport across the network is almost always TCP. TCP sends data in "windows". A window defines the maximum amount of data a sender can transmit before receiving acknowledgement from the receiver. Since it takes a round-trip time to receive the acknowledgement from the receiver the maximum throughput is the amount of data in a window divided by the round trip time. TCP slow start and congestion control features designed to increase reliability also make the throughput problem worse.

Application chattiness multiplies the effect of latency: On top of TCP, applications have their own communications protocols. For example, Microsoft Windows uses CIFS, the Common Internet File System. Microsoft Exchange uses

MAPI, the Messaging Application Programming Interface protocol. Web based applications rely on HTTP, and so forth.

Some protocols (application or transport) are extremely "chatty", which means they generate hundreds or thousands of round trips from client to server, even to accomplish seemingly simple tasks. For example, dragging and dropping a 1 MB file in Windows can trigger over 4,000 WAN round trips. On a LAN, when the latency between client and server is often less than a tenth of a millisecond, those thousands of round trips are completed virtually instantaneously. When the same operation is done on a WAN, the latency is usually in the range of 50 ms to 250 ms, or even more when satellites are involved.

Application protocols also have a limited amount of data they can transmit on each round trip. So the problem of many round trips is worse for large files. If the application protocol has a "transfer size" of 16 KB, then 16 MB file will require 1,000 trips, just to deliver the data, plus lots of additional round trips generated by the application to manage the data transfer, file system operations, or whatever other operations are required.

A similar chattiness issue applies to the TCP layer, which affects web-based business apps, as well as applications like Notes, FTP and other mission critical applications.

Exposing The Myths

Over the past few years, vendors have created a number of products to accelerate application performance. These solutions, often categorized as WAN Optimisation or WAFS, have fallen into three categories:

1. TCP optimisation, 2. Compression 3. Caching

IT professionals have learned that these solutions are either insufficient to address performance across a wide range of applications, introduce additional complexity, or both. While each of these solutions can solve specific issues with application performance across the WAN, the myth still exists that they are general solutions to application performance.

Myth #1: You can solve application performance with TCP optimisation alone

Many IT professionals are aware that TCP as originally defined has a maximum window size of 64 KB (the typical amount of data that can be carried in each TCP round-trip), and that the limit can be modified with some work. In many cases, the configured maximum TCP window size is even smaller - 16 KB or 32 KB, which makes the problem even worse. Even companies who elect to go the route of modifying TCP find that fixing or improving TCP does not help application performance if the application protocol is less efficient than TCP.

Benefits Of Consolidation Of Remote Site Infrastructure

- Reduces cost and complexity
- Improves compliance
- Improves data and network security
- Improves resource utilisation
- Eliminates need for costly WAN bandwidth upgrades
- Eliminates write consistency issues associated with caching
- Frees up WAN capacity for VoIP and video applications

Hardware solutions exist to modify TCP's behavior in other ways across the WAN to increase its throughput, but modifying the TCP layer in the networking stack does nothing to improve performance issues caused by higher layer protocols. For many applications like Windows file sharing or Exchange, the application protocol (CIFS and MAPI, respectively) are much chattier and less efficient than TCP itself. Thus, making TCP more efficient can be helpful, but in many cases this approach alone is insufficient.

Myth #2: You can solve application performance with compression

Companies that attribute application performance issues to lack of bandwidth often reach the conclusion that they can solve the problem by adding compression appliances. This is equivalent to adding more bandwidth. More bandwidth is helpful, but again it's insufficient to solve the problem. Adding bandwidth does not help alleviate the chattiness of the application, which means that all those round trips still have to take place. No matter how much bandwidth you buy, once the initial congestion has been alleviated, the application performance will not be materially affected.

Myth #3: You can solve application performance with caching

Some companies have investigated caching appliances as a way to enable site consolidation. That approach can work for single data types, but will not provide a general solution and often is used just to hide the underlying performance problem. For Exchange, there are special purpose mail caching appliances available, but they are not a general purpose solution to the problem. Caching is an application-specific technology: File caching works for file systems, Web caching works for web pages, mail caching works for email, and so on. So while adding an Exchange mail cache will help by storing attachments locally, it adds complexity and only affects the perceived performance of Exchange.

Another issue for file caching is write consistency. Often caching products will implement elaborate file locking mechanisms to prevent two users from writing the same file. But in the event of network outages and/or box failure, these mechanisms can fail leading to catastrophic results.

With Exchange 2003 and Outlook 2003 Microsoft introduced integrated client-side caching to address performance across the WAN. This hides the delay in getting email from servers to clients by not displaying any new email headers until the entire email and any attachments are fully delivered. Client-side caching can improve the perceived user experience but may cause much heavier traffic across the WAN, since emails are delivered to the client that might have been deleted without being read if only the headers were seen. ■